



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to PGM PETROL BUNK

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

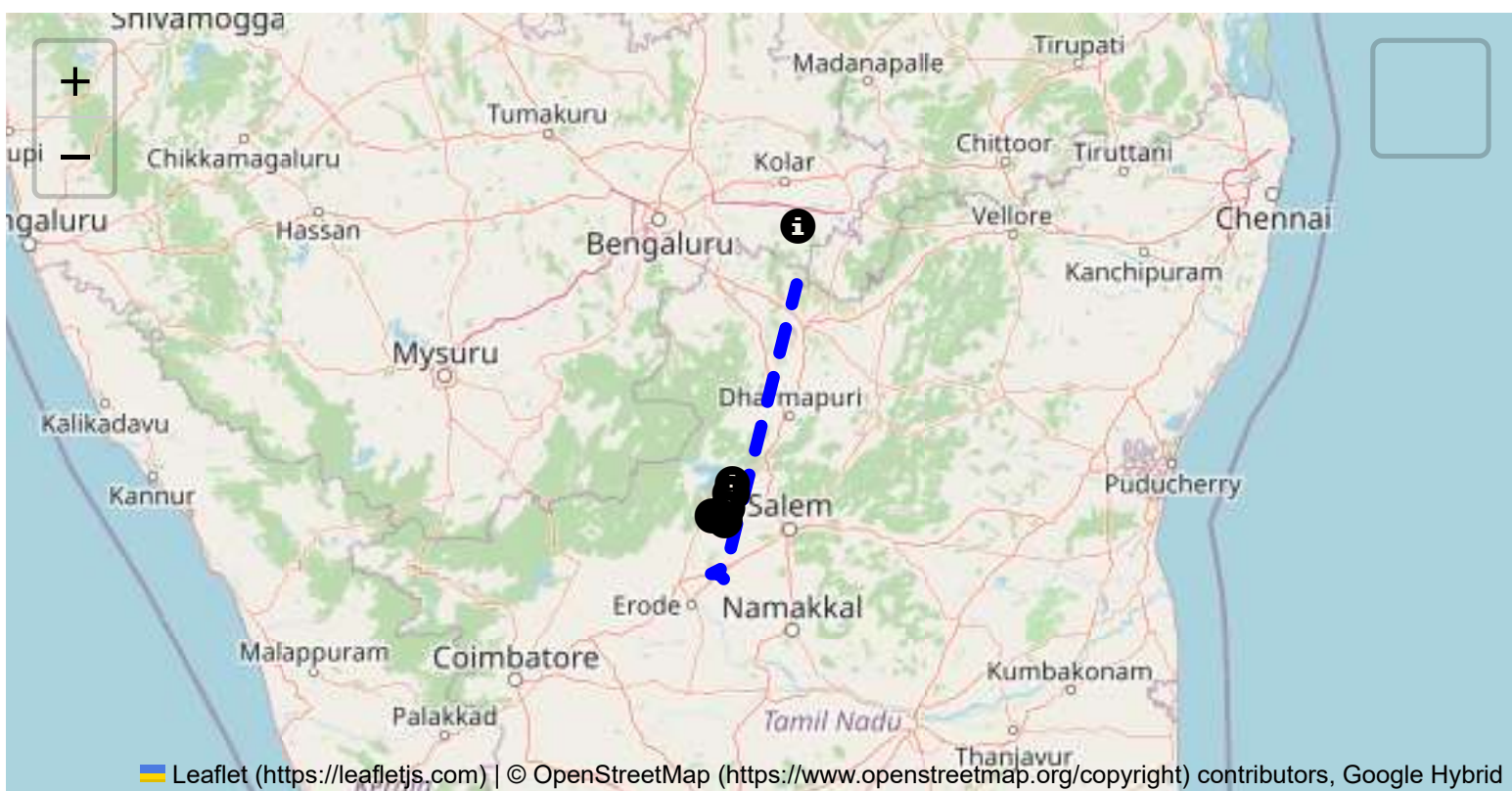
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

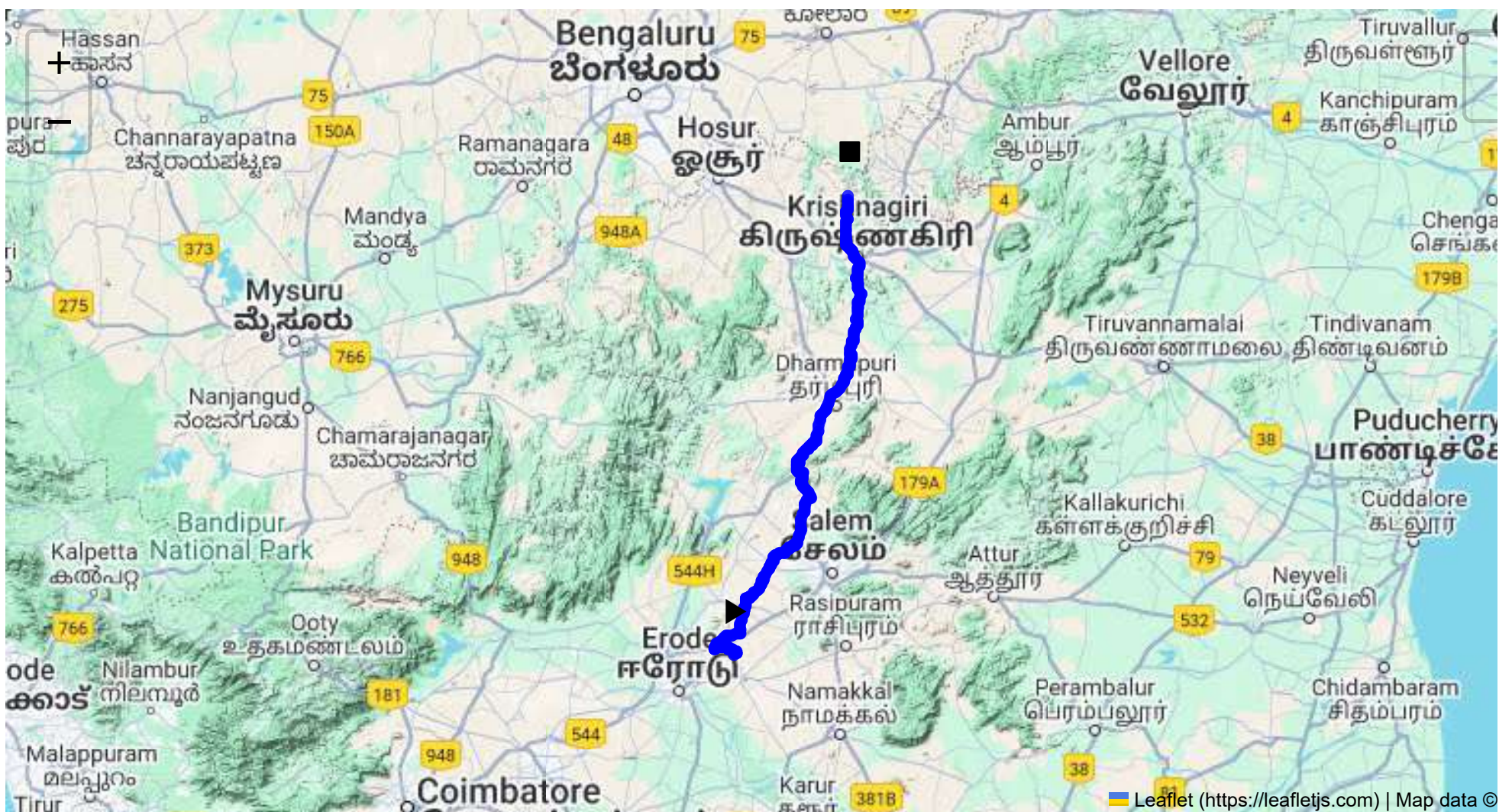
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 172.27 km
Estimated Duration: 3.3 hours
Adjusted Duration (Heavy Vehicle): 4.2 hours
Start: (11.4381, 77.8734)
End: (12.693096, 78.191242)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to M5VR+8G5, Veppanapalli is approximately 172.27 kilometers long and traverses primarily through Tamil Nadu's rural and semi-urban areas. Key waypoints include sections along the Sankari RS - Erode Privu Road, Bhavani Main Road, and SH 86, which are main thoroughfares connecting various towns.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences hot and dry conditions, with a monsoon season contributing to occasional heavy rains, especially between October and December. These rains could lead to slippery roads and minor flooding, particularly in poorly drained areas.

3. Analysis of Traffic Patterns

Traffic density is generally higher around urbanized areas such as Sangagiri and along main roads like the Bhavani Main Road. Peak hours typically occur in the morning (8 AM - 10 AM) and evening (6 PM - 8 PM). Expect congestion near market areas and school zones.

4. Assessment of Road Quality and Infrastructure

The roads along the route vary in quality. Major highways like SH 86 are generally well-maintained but secondary roads may have potholes or uneven surfaces. Quick lane changes are possible in some areas due to narrow lanes, and construction work can occasionally disrupt flow.

5. Suggestions for Alternative Routes for Emergencies

For emergencies, an alternative route could involve taking the NH44 heading north from Sangagiri, which can bypass localized roadblocks or traffic jams along smaller roads. Keeping updated maps ready with live traffic information would be beneficial.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu enforces strict regulations on the transport of hazardous materials, including designated times and routes for such vehicles. Ensure all driver documentation is up to date and adherence to hazardous materials transport guidelines like proper signage and documentation is strictly observed.

7. Overview of Historical Incidents

There have been occasional incidents involving heavy goods vehicles due to brake failures or spills, generally in high-traffic or steep gradient areas. Awareness and adherence to state-mandated vehicle checks and speed limits mitigate these risks.

8. Environmental Considerations and Sensitive Areas

The route passes through agricultural and semi-rural terrain. Care should be taken to prevent contamination, particularly near water bodies and inhabited zones, to protect local biodiversity and communities.

9. Analysis of Communication Coverage

While telecommunications coverage is generally good along SH 86 and other major roads, pockets of weak signal strength can be expected in more rural stretches, notably between smaller villages. Carrying a satellite phone or a radio for emergency communication is advised.

10. Estimated Emergency Response Times

Response times vary, with quicker access (within 30 minutes) near towns with established emergency services like Sangagiri and Sankari. Rural segments might expect delays up to 60-90 minutes due to distance from centralized emergency hubs.

11. Overall Summary of Risk Assessment

This route contains a mix of well-maintained highways and a few challenging rural road segments. Weather, congestion, and road quality present typical operational risks, but they can be mitigated by adhering to route-specific regulations and maintaining real-time communication for dynamic route adjustments. Regular maintenance checks and cautious driving practices are advised for safe transport of hazardous materials across this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	High	11.45571, 77.81702	15 KM/Hr
7	Turn	Medium	11.45654, 77.81706	30 KM/Hr
8	Turn	Medium	11.46916, 77.82431	30 KM/Hr
9	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
10	Turn	High	11.49465, 77.88488	15 KM/Hr
11	Turn	Medium	11.56096, 77.89931	30 KM/Hr
12	Turn	Medium	11.57649, 77.90610	30 KM/Hr
13	Turn	Medium	11.59214, 77.90656	30 KM/Hr
14	Turn	Medium	11.63833, 77.94818	30 KM/Hr
15	Turn	High	11.64243, 77.95198	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
16	Turn	Medium	11.64721, 77.95118	30 KM/Hr
17	Turn	Medium	11.67870, 77.96846	30 KM/Hr
18	Turn	High	11.70227, 77.97864	15 KM/Hr
19	Turn	Medium	11.71416, 77.99604	30 KM/Hr
20	Turn	Medium	11.71715, 77.99969	30 KM/Hr
21	Turn	Medium	11.71921, 78.01133	30 KM/Hr
22	Turn	Medium	11.72737, 78.02248	30 KM/Hr
23	Turn	High	11.73775, 78.03465	15 KM/Hr
24	Turn	High	11.73651, 78.03878	15 KM/Hr
25	Turn	High	11.74189, 78.04660	15 KM/Hr
26	Turn	Medium	11.77961, 78.05775	30 KM/Hr
27	Turn	Medium	11.86873, 78.08374	30 KM/Hr
28	Turn	Medium	11.87966, 78.08187	30 KM/Hr
29	Turn	Medium	11.92609, 78.06513	30 KM/Hr
30	Turn	Medium	11.93149, 78.06485	30 KM/Hr
31	Turn	Medium	11.93455, 78.06295	30 KM/Hr
32	Turn	Medium	11.93719, 78.05372	30 KM/Hr
33	Turn	Medium	11.95457, 78.04685	30 KM/Hr
34	Turn	Medium	11.95688, 78.04803	30 KM/Hr
35	Turn	High	11.95828, 78.04983	15 KM/Hr
36	Turn	Medium	11.95997, 78.04946	30 KM/Hr
37	Turn	Medium	11.96335, 78.05141	30 KM/Hr
38	Turn	High	11.96408, 78.05131	15 KM/Hr
39	Turn	Medium	11.96511, 78.04966	30 KM/Hr
40	Turn	High	11.96608, 78.04919	15 KM/Hr
41	Turn	Medium	11.96819, 78.05088	30 KM/Hr
42	Turn	Medium	12.17005, 78.16792	30 KM/Hr
43	Turn	Medium	12.19919, 78.18172	30 KM/Hr
44	Turn	Medium	12.22472, 78.19140	30 KM/Hr
45	Turn	Medium	12.24505, 78.19496	30 KM/Hr
46	Turn	Medium	12.25078, 78.19988	30 KM/Hr
47	Turn	Medium	12.25370, 78.20034	30 KM/Hr
48	Turn	Medium	12.29150, 78.19930	30 KM/Hr
49	Turn	Medium	12.39930, 78.21762	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
50	Turn	Medium	12.40604, 78.22285	30 KM/Hr
51	Turn	Medium	12.42500, 78.22672	30 KM/Hr
52	Turn	Medium	12.50188, 78.22338	30 KM/Hr
53	Turn	Medium	12.50598, 78.22301	30 KM/Hr
54	Turn	High	12.56124, 78.18177	15 KM/Hr
55	Turn	High	12.56666, 78.18432	15 KM/Hr
56	Turn	Medium	12.56740, 78.18283	30 KM/Hr
57	Turn	Medium	12.66056, 78.18940	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
10	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
12	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
13	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
14	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
15	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
16	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
19	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
23	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
25	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
26	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
27	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
28	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
33	clinic	Shri Narayana Bone & Joint Clinic	12.523617, 78.210613	30 km/h	Medium
34	clinic	AZ Family Clinic	12.5234942, 78.2127422	30 km/h	Medium
36	hospital	KVS Hospital	12.5277857, 78.2055049	30 km/h	Medium
37	hospital	Ga Heart Hospital	12.5287708, 78.2055336	30 km/h	Medium
38	clinic	Psychiatric Centre	12.529476, 78.205064	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
11	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
22	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
24	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
29	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
30	school	Nalanda Metric Higher Secondary School	12.488407, 78.21856	30 km/h	Medium
31	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
32	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
35	school	Vivekananda School	12.5231937, 78.2121429	30 km/h	Medium
39	school	Sri Saraswathi Vidyalaya	12.5620799, 78.183049	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
40	school	Shri Maruth Matric Hr Sec School	12.6545021, 78.1858443	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45654, 77.81706



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46916, 77.82431



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56096, 77.89931



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57649, 77.90610



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59214, 77.90656



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63833, 77.94818



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64243, 77.95198



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64721, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67870, 77.96846



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70227, 77.97864



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71416, 77.99604



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71715, 77.99969



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71921, 78.01133



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72737, 78.02248



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73775, 78.03465



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03878



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.74189, 78.04660



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77961, 78.05775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86873, 78.08374



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87966, 78.08187



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92609, 78.06513



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93149, 78.06485



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93455, 78.06295



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93719, 78.05372



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95457, 78.04685



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95688, 78.04803



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95828, 78.04983



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95997, 78.04946



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96335, 78.05141



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96408, 78.05131



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96511, 78.04966



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96608, 78.04919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96819, 78.05088



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17005, 78.16792



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.19919, 78.18172



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.22472, 78.19140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24505, 78.19496



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25078, 78.19988



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25370, 78.20034



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.29150, 78.19930



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.39930, 78.21762



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.40604, 78.22285



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.42500, 78.22672



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.50188, 78.22338



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.50598, 78.22301



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.56124, 78.18177



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.66056, 78.18940
